

The Vision Node

Phase 2 — a real-time perception layer for the VillaOS platform

Project	VillaOS — AI + IoT + Robotics smart-villa platform
Document	Phase 2 project report (public)
Phase	2 of 6 — Computer vision
Status	Core complete

EXECUTIVE SUMMARY

Phase 2 gives VillaOS sight. It introduces a real-time **perception node** that watches a camera feed, understands what it sees — from motion, to faces, to *known versus unknown identities* — and publishes those observations onto the platform's shared message bus for any other node to act on. The result is not a face-recognition demo; it is a production-shaped perception service that composes with the rest of the platform through the same architectural grammar as every other VillaOS component.

1. Context

VillaOS is a multi-phase engineering platform built around one principle: don't build disconnected demos, build one system that grows deeper at every stage.

Phase 1 established a device that could be *commanded* and could *report its own state* over a message bus. Phase 2 adds the sense that turns a smart home into an aware one: a node that *observes the world* and reports what it sees onto that same bus, in the same language — so perception and control compose without either side knowing the other's internals.

2. What the vision node does

The node performs a layered chain of perception, each layer more specific than the last:

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Motion awareness — detects meaningful movement while ignoring noise, gradual lighting change, and shadows.

- 2 **Face detection** — locates human faces in the frame, robust to head angle, tilt, and varied lighting.
- 3 **Face recognition** — distinguishes *known* individuals from *unknown* ones, with a confidence score for each identification.
- 4 **Event publishing** — emits every observation onto the platform bus as a structured, timestamped event other nodes can react to.

Because these observations flow onto a shared bus, the platform composes behaviours across independent nodes — for example, a lighting response to an unrecognised visitor — without either node being aware of the other.

3. Architecture

The node is built around a single idea: **one processing loop that grows more capable at each stage**. A frame is captured, understood, and the observation is published — and "understood" deepens layer by layer, from *something moved*, to *that is a face*, to *that is a known person*. Publication connects the node to the platform through a publish/subscribe bus, using a strict separation between **intent** and **observed reality** borrowed from control-systems design.

Property	What it means
Composable	The node is a peer on the bus, not a monolith. New behaviours emerge from combining nodes, not rewriting them.
Extensible	Adding a recognised identity requires no retraining — enrolment is additive by design.
Forward-compatible	The publish model maps directly onto robotics middleware, positioning VillaOS for its later autonomous-robotics phases.

4. Technology

Layer	Approach
Capture	Networked camera stream over the local network
Motion	Adaptive background-subtraction
Face detection	Deep-learning detector with facial-landmark output
Face recognition	Neural feature embeddings compared by similarity
Messaging	Lightweight publish/subscribe message broker

The recognition approach represents each face as a compact numerical embedding — a learned "fingerprint" — and identifies people by measuring similarity in that representation space. It is the same class of technique that underpins modern image search and recommendation systems, chosen for its accuracy, its resilience to real-world lighting and pose, and its clean scaling: new identities are added by storing one more reference, never by retraining a model.

5. Engineering principles

BUILD VERTICALLY

Each capability was taken end to end — raw frame to published event — before the next was started. No half-finished horizontal layers that teach nothing and demo as nothing.

PRIVACY BY DESIGN

All biometric data is treated as sensitive: generated and stored locally, never transmitted off-device, and explicitly excluded from version control. Personal data stays on the owner's own hardware.

PLATFORM OVER DEMO

Every capability integrates into the shared ecosystem rather than standing alone, so the whole compounds into a coherent system instead of a folder of scripts.

TUNABLE POLICY

Recognition confidence is governed by an adjustable acceptance threshold — the single knob that expresses the security posture, trading false accepts against missed matches.

6. Results

The node runs a complete perception pipeline in real time on commodity hardware: from live camera frames through detection, alignment, and recognition, to a throttled stream of structured observation events on the platform bus. Recognition holds reliably across the head angles and lighting conditions where classical methods fail, and integrates live with independently-built platform nodes over a shared message bus — the moment the system stops being a collection of parts and starts behaving as a platform.

7. Roadmap

Phase 2 continues and hands off into the robotics phases of VillaOS:

Automatic number-plate recognition

Object detection & visitor analytics

Edge-inference optimisation

Robotics-middleware migration

The final step promotes the perception node into a first-class node in a full robotics stack — the foundation for VillaOS's autonomous-systems work.

VillaOS is an independent, multi-phase engineering platform. This document describes the system's design and capabilities at a high level; implementation detail is maintained privately.